

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSEAI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Architectural Modelling (CO2)

Group No : 4

Team Members :

Roll No	Student Name
AMSC.U4AIE25007	Rohan S
AM.SC.U4AIE25001	Abhijith R Pillai
AM.SC.U4AIE25031	Adarsh V
AM.SC.U4AIE25052	Gautam Dileep

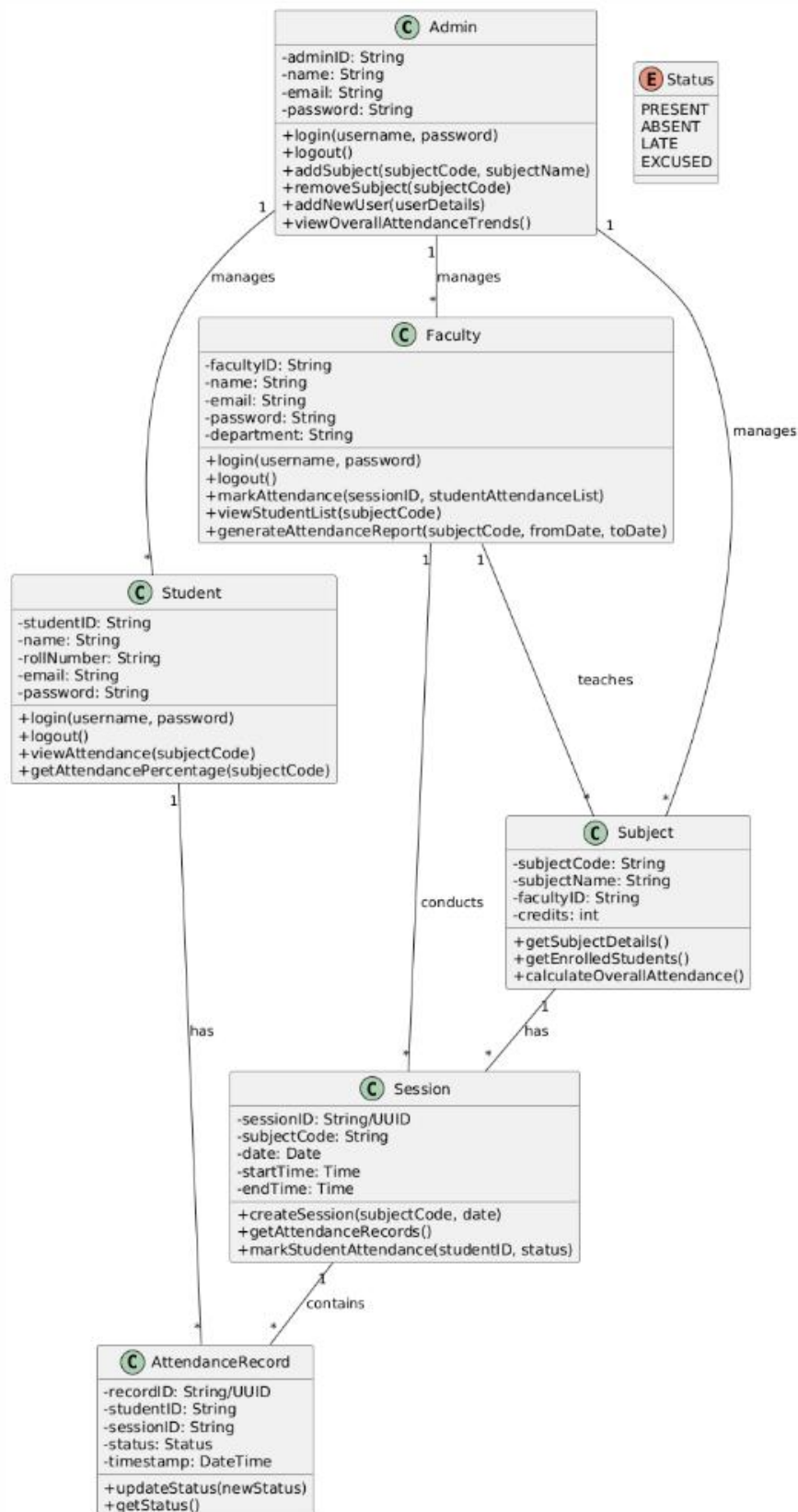
1. Problem Title :

Student Attendance Management System

STRUCTURAL DESIGN

2. Class diagram

The class diagram represents the static structure of the system showing classes, attributes, methods and the relationships between the objects.



BEHAVIUIORAL DESIGN

3. Use-case diagram

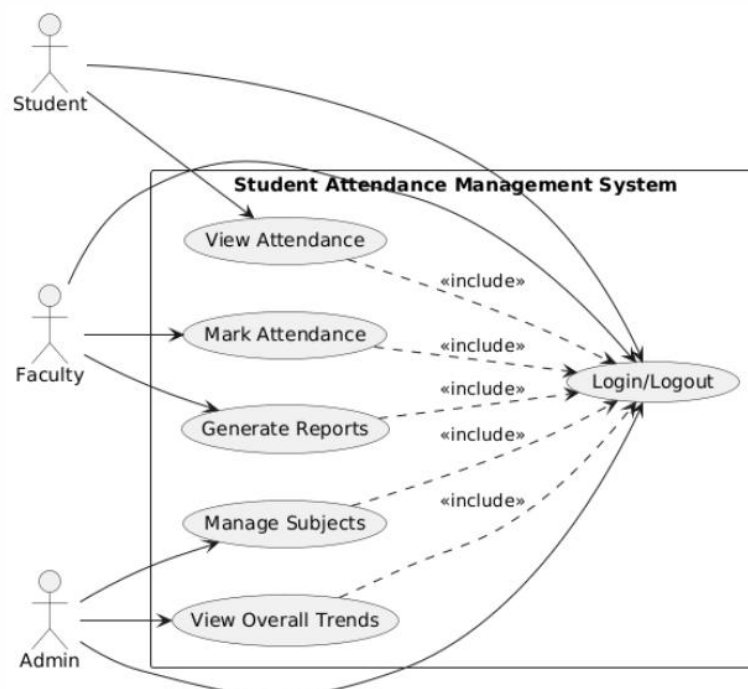
a. List of actors

1. **Student:** A user who logs in to check their specific attendance records and percentages for subjects.
2. **Faculty:** A teacher responsible for conducting classes and marking student presence or absence.
3. **Admin:** The system overseer who manages subjects and users without marking attendance.

b. List of use-cases

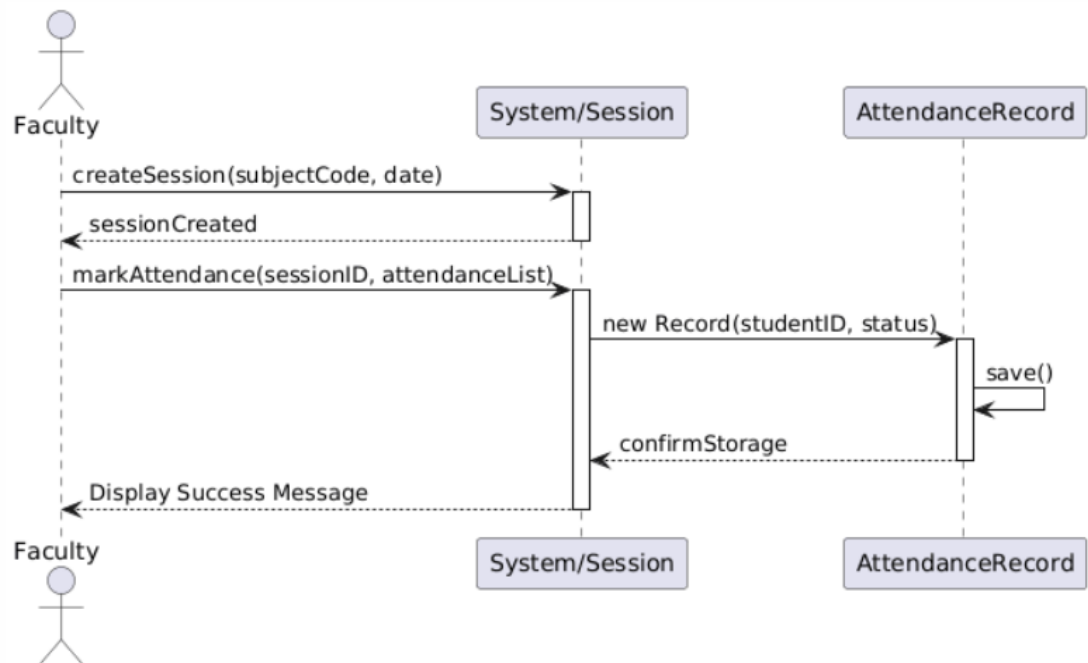
1. **Login/Logout:** Authenticates all users (Student, Faculty, Admin) to grant secure access.
2. **Mark Attendance:** Allows Faculty to record student statuses (Present/Absent) for a specific session.
3. **View Attendance:** Enables Students to check their own attendance history and percentages.
4. **Manage Subjects:** Provides the Admin with tools to add or remove subjects from the system.
5. **Generate Reports:** Allows Faculty to create detailed attendance summaries for administrative use.
6. **View Overall Trends:** Provides Admin with high-level statistics of attendance across all courses.

c. Use case diagram:



4. Sequence diagram :

- a. **Faculty Marking Attendance:** The Faculty initiates the process by selecting a specific subject and creating a class session. Once the session is active, the Faculty submits the attendance status for the student list. The system then instantiates individual Attendance Record objects for each student and saves them to the database, returning a success message upon completion.



- b. **Student Viewing Percentage:** The Student logs into the portal and requests their attendance percentage for a particular subject. The System acts as an intermediary, calling the calculation logic within the Subject class. The Subject class aggregates all relevant records to compute the final ratio, which is then passed back to the Student's interface.

